ASSIGNMENT- 8 DBMS(MA 2205)

Student Table

student_id	name	dept_name	tot_cred
1	Alice	Computer Science	120
2	Bob	Mathematics	90
3	Charlie	Physics	100

Course Table

course_id	title	dept_name	credits
CS101	Intro to CS	Computer Science	4
MATH101	Calculus I	Mathematics	3
PHYS101	Physics I	Physics	4

Takes Table

student_id	course_id	semester	year	grade
1	CS101	Fall	2024	A
2	MATH101	Spring	2024	B
3	PHYS101	Fall	2024	A

Instructor Table

instructor_id	name	dept_name	salary
101	Dr. Smith	Computer Science	90000.00
102	Dr. Johnson	Mathematics	85000.00
103	Dr. Lee	Physics	87000.00

Teaches Table

instructor_id	course_id	semester	year
101	CS101	Fall	2024
102	MATH101	Spring	2024
103	PHYS101	Fall	2024

Questions

1. **HAVING Clause:**
   Write a query to find all departments where the average credits of courses offered is more than 3. Display the department name and the average credits.
2. **EXISTS Clause:**
   Find all students who have taken all courses offered by the 'Computer Science' department.
3. **ALL Clause:**
   Find the names of instructors who have a higher salary than all instructors in the 'Biology' department.
4. **Combination (HAVING and EXISTS):**
   List all courses that have been taken by all students in the 'Computer Science' department. Display the course ID and title.
5. **Nested Subquery with EXISTS:**
   Find all students who have taken at least one course with every instructor in their department.